

Chapter 07

Planar Architectures and Through-Substrate Thermal Design

From Laboratory Cylinders to Production-Ready Via Arrays

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

- LO-07.1** Translate cylindrical transformation devices into Cartesian (planar) geometries and quantify the resulting edge effects using the cloaking efficiency metric R_{cloak} .
- LO-07.2** Design thermal switch plates with binary and continuous gradient functionality for wafer handling applications, including non-ideal performance limits.
- LO-07.3** Derive the effective medium expressions for via arrays (TSV, TGV) as engineered anisotropic thermal metamaterials, and state their validity boundaries.
- LO-07.4** Apply four via array design patterns—uniform, clustered, graded, and annular—to distinct thermal management problems using the pattern selection decision table.
- LO-07.5** Calculate thermal conductance through glass substrates with TGV arrays and apply the glass-specific design rules (DR-09.1 through DR-09.9).
- LO-07.6** Perform thermal-electrical co-design for electrostatic chuck integration with metamaterial vias, including keep-out zone penalty calculations.
- LO-07.7** Integrate via array architectures into the Mode-Path Matrix framework by identifying via arrays as high-conductance network edges.
- LO-07.8** Produce a complete via-array prescription table as the standard design output for any conduction-dominated substrate thermal path.

Abbreviations and Nomenclature

Abbreviation	Meaning
AR	Aspect Ratio (of vias)
BGA	Ball Grid Array
CMP	Chemical Mechanical Planarization
CTE	Coefficient of Thermal Expansion
CVD	Chemical Vapor Deposition
DRIE	Deep Reactive Ion Etching
EMT	Effective Medium Theory

Abbreviation	Meaning
ESC	Electrostatic Chuck
FDM	Finite Difference Method
FEA	Finite Element Analysis
GRIN	Gradient Index
KOZ	Keep-Out Zone
MG	Maxwell–Garnett
MPM	Mode-Path Matrix
PVD	Physical Vapor Deposition
SiP	System in Package
TGV	Through-Glass Via
TIM	Thermal Interface Material
TSV	Through-Silicon Via

Symbol	Meaning	SI Unit
κ	Thermal conductivity tensor	$\text{W m}^{-1} \text{K}^{-1}$
$\kappa_{\parallel}$	Parallel (in-plane) conductivity	$\text{W m}^{-1} \text{K}^{-1}$
$\kappa_{\perp}$	Perpendicular (cross-plane) conductivity	$\text{W m}^{-1} \text{K}^{-1}$
κ_{via}	Via fill material conductivity	$\text{W m}^{-1} \text{K}^{-1}$
κ_{matrix}	Matrix (substrate) conductivity	$\text{W m}^{-1} \text{K}^{-1}$
κ_{eff}	Effective conductivity	$\text{W m}^{-1} \text{K}^{-1}$
f_{via}	Via fill fraction (area)	—
d_{via}	Via diameter	μm
p_{via}	Via pitch	μm
t_{sub}	Substrate thickness	μm
AR	Via aspect ratio ($t_{\text{sub}}/d_{\text{via}}$)	—
Γ	Mode dominance ratio ($G_{\text{rad}}/G_{\text{cond}}$)	—
G_{cond}	Conduction conductance	W K^{-1}
G_{rad}	Radiation conductance	W K^{-1}
G	Thermal conductance	W K^{-1}
R_{th}	Thermal resistance	K W^{-1}
R_{cloak}	Cloaking efficiency (0 to 1)	—
β	Bottleneck ratio	—
M	Conductance mismatch ratio	—
T	Temperature	K
A	Area	m^2
L	Characteristic length	m

Subscript/Superscript Conventions

Notation	Convention
cond, rad	Conduction and radiation modes, respectively
eff	Effective (homogenized or composite) value
$\parallel, \perp$	Parallel and perpendicular to substrate plane
via	Via fill material
mat	Matrix (substrate) material

sub	Substrate
th	Thermal (as in τ_{th} , thermal time constant)
proc	Process (as in τ_{proc} , allowable process time)

Anisotropy convention for via arrays (this chapter): $\kappa_{\perp}^{eff}/\kappa_{\parallel}^{eff}$ (through-plane over in-plane), which is the ratio relevant to through-substrate heat transfer. Values $\gg 1$ indicate strong preferential heat flow through the substrate. This differs from the Chapter 5 convention ($\kappa_{\parallel}/\kappa_{\perp}$) used for in-plane metamaterial shells; the two are reciprocals.

Design rules for this chapter follow the DR-09.n numbering convention (inherited from v2.2 Chapter 9 for continuity). Topological design rules inherited from Chapter 4 retain the DR-T.n prefix. Worked examples use WE-07.n. Learning objectives use LO-07.n.

Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

A complete via array thermal design methodology—from effective medium equations through design pattern selection to production-ready specifications—integrated into the Mode-Path Matrix framework, with a standard prescription table as the output format.

To help you navigate, Figure 07.1 shows the chapter “spine” (workflow) and how each section feeds into the final via array design toolkit.

Figure 07.1: Chapter 07 Roadmap

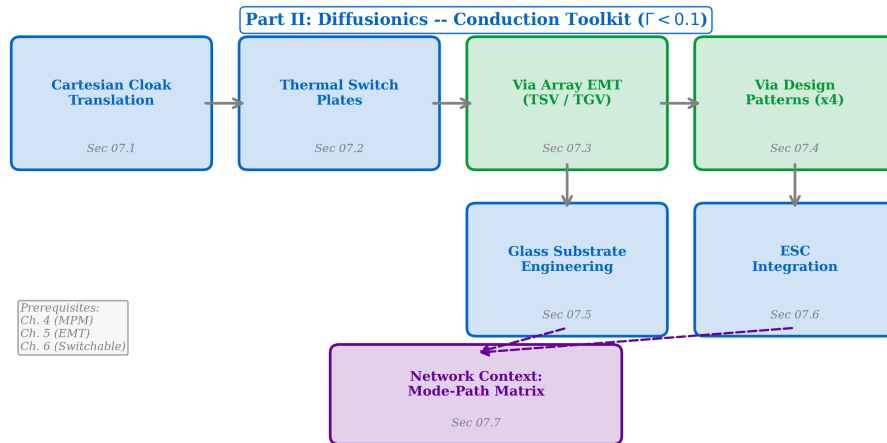


Figure 07.1: Chapter 07 roadmap. The narrative spine translates cylindrical metamaterial concepts (§07.1) through planar switch plates (§07.2) into via array effective media (§07.3–§07.4), with specialization to glass substrates (§07.5) and ESC integration (§07.6). The network context (§07.7) ties all architectures back to the Mode-Path Matrix. Blue boxes indicate conduction-dominated sections ($\Gamma < 0.1$). The MPM-first design pipeline (Reader Recipe) anchors the entire workflow.

Reader Recipe

Via Array Design Pipeline — MPM-First Approach

Every via array design in this chapter follows a single pipeline, anchored by the Mode-Path Matrix (Chapter 4). Use this checklist as your “fast start”; each step references the relevant section for details.

- Step 1: Start from MPM (§07.7):** identify the critical path; confirm the substrate is the bottleneck ($\beta > 0.5$).
- Step 2: Compute Γ :** confirm $\Gamma < 0.1$ (conduction tools apply). If $\Gamma > 0.1$, radiative coupling matters—see Chapter 9.
- Step 3: Select pattern** from Table 07.7 (§07.4) based on bottleneck type (uniform, hotspot, concentrator, or cloak).
- Step 4: Synthesize layout:** compute f_{via} , d_{via} , p_{via} from EMT equations (§07.3); verify EMT validity (Table 07.5).
- Step 5: Reliability gate:** check DR-09.2 through DR-09.9 (aspect ratio, wall thickness, CTE stress, KOZ, edge spacing, temperature limits).
- Step 6: Verify:** update network with G_{via} ; recalculate critical path (DR-T.6). If the substrate is no longer the bottleneck, the design is complete.
- Step 7: Output:** fill in the *Via-Array Prescription Table* (Table 07.6). This is your deliverable.

Quick navigation: If you need a fast via design, jump to §07.3–§07.5 and WE-07.1. If you need cloaking translation, start with §07.1 + Pattern 4 (§07.4.4). If you are doing system-level integration, start with §07.7 and work backward through the pipeline.

Pain Point:

The elegant cylindrical cloaks and concentrators of Chapter 5 are beautiful physics—but semiconductor wafers are flat, substrates are rectangular, and fabs run Cartesian processes. Translating cylindrical transformation devices into planar architectures introduces edge effects, requires Cartesian coordinate mappings, and demands compatibility with standard lithographic patterning. Worse, the binary layered composites of Chapter 5 achieve anisotropy ratios of $\sim 15:1$ at best in production—whereas via arrays, already ubiquitous in semiconductor packaging, routinely achieve 10:1 to 100:1 using standard CMOS and glass-substrate processes. **Via arrays are the production thermal metamaterial**, yet no existing textbook treats them as such. This chapter bridges the gap between metamaterial theory and fabrication reality.

Continuity: where Chapter 7 sits in the book. Part II (Chapters 5–7) is the *Diffusionics* toolkit for conduction-dominated paths ($\Gamma < 0.1$) identified by the Mode-Path Matrix (Chapter 4). Chapter 5 established transformation thermotics and effective media—how to *prescribe* an anisotropic conductivity tensor κ using coordinate transformations and binary composites. Chapter 6 introduced switchable and asymmetric devices—how to *change* conductance in time via doublets, liquid metals, and thermal diodes. Chapter 7 closes Part II by answering the production question: *how do we implement those tensors on planar, Cartesian substrates using processes that fabs already run?* Binary layered composites (Chapter 5) cap effective anisotropy at $\kappa_{\parallel}/\kappa_{\perp} \sim 15:1$ in production; via arrays break this barrier, reaching $\kappa_{\perp}^{\text{eff}}/\kappa_{\parallel}^{\text{eff}} \sim 10:1$ to 100:1

using a single lithographic step.

Promise: what this chapter delivers. By the end of this chapter, the reader will be able to: (i) translate cylindrical metamaterial designs into Cartesian geometries and quantify the degradation using a formal cloaking efficiency metric R_{cloak} and the “Cartesian tax” (DR-09.10); (ii) design thermal switch plates with realistic engineering constraints including radiation leakage, contact resistance, and cycling reliability; (iii) use effective medium theory (EMT) to convert via array layouts into thermal conductivity specifications, with explicit validity boundaries and a model fidelity ladder; (iv) select the optimal via pattern from a decision table keyed to bottleneck type; and (v) integrate the resulting design into the Mode-Path Matrix as a high-conductance network edge, producing a standard *via-array prescription table* as the final output. The central message is that TSV/TGV via arrays are not merely interconnects—they are the *production thermal metamaterial*: a manufacturable route to extreme effective anisotropy, plus a pattern library (uniform/clustered/graded/annular) that maps directly onto real packaging needs.

How to use this chapter. If you need a quick via design for a known thermal bottleneck, jump to the Reader Recipe above and follow the seven-step pipeline, using §07.3–§07.5 and WE-07.1. If you need to translate a cylindrical cloak or concentrator into a planar implementation, start with §07.1 and Pattern 4 (§07.4.4). If you are performing system-level thermal optimization, begin with §07.7 (network context) and work backward through the pipeline. **No Diffusionics design is complete until it has been translated into a planar, Cartesian, via-array prescription that passes all DR-09 gates.** This chapter is the Part II exit gate.

Forward link. The via-array toolkit developed here returns in Chapter 11 (wafer/chuck thermal control) and Chapter 13 (manufacturing and metrology), where design rules become yield and QC requirements. The prescription table format (Table 07.6) is designed for compatibility with the automated design workflow of Chapter 15.

07.1 Cartesian Cloak Translation

The thermal cloaks designed in Chapter 5 via cylindrical coordinate transformations produce annular shell geometries. Real semiconductor substrates, wafer chucks, and interposers are planar (Cartesian). This section develops the mapping from cylindrical to Cartesian metamaterial architectures and introduces a formal performance metric to quantify the translation fidelity.

07.1.1 Cylindrical-to-Cartesian Mapping

Consider a cylindrical thermal cloak of inner radius R_1 and outer radius R_2 , with the radial and azimuthal conductivities prescribed by Eq. (05.3):

$$\kappa_r = \kappa_0 \frac{r - R_1}{r}, \quad \kappa_\theta = \kappa_0 \frac{r}{r - R_1} \quad (07.1)$$

where r is the radial coordinate in the transformed space. To implement this in a rectangular substrate, we employ a Cartesian approximation by mapping the annular shell to a rectangular frame.

The simplest Cartesian cloak uses a coordinate squeezing in the x -direction:

$$x' = \frac{b-a}{b} x + a \operatorname{sgn}(x), \quad y' = y, \quad z' = z \quad (07.2)$$

where a is the half-width of the cloaked region and b is the half-width of the outer metamaterial frame. The resulting conductivity tensor in the Cartesian cloak shell is:

$$\boldsymbol{\kappa}' = \frac{\kappa_0}{J_{11}} \begin{pmatrix} J_{11}^2 & 0 \\ 0 & 1 \end{pmatrix} = \kappa_0 \begin{pmatrix} (b-a)/b & 0 \\ 0 & b/(b-a) \end{pmatrix} \quad (07.3)$$

where $J_{11} = (b - a)/b$ is the Jacobian element. The required anisotropy ratio is:

$$\frac{\kappa_{\parallel}}{\kappa_{\perp}} = \left(\frac{b}{b - a} \right)^2 \quad (07.4)$$

For $a/b = 0.3$ (cloaking 30% of the frame width), $\kappa_{\parallel}/\kappa_{\perp} = (1/0.7)^2 \approx 2.04$ —easily achievable with any material pair from Table 05.5. For $a/b = 0.5$, $\kappa_{\parallel}/\kappa_{\perp} = 4.0$; for $a/b = 0.7$, $\kappa_{\parallel}/\kappa_{\perp} = 11.1$, requiring higher-contrast pairs or via arrays.

Conduction Mode (Diffusionics)

Key advantage of Cartesian cloaks: The anisotropy is uniform throughout the shell (Eq. 07.3), unlike the spatially varying $\kappa_r(r)$ and $\kappa_{\theta}(r)$ of the cylindrical cloak. This means a *single* binary composite or via array density can implement the entire cloak—no graded layers are needed.

Cloaking efficiency metric. To quantify translation fidelity, we use the cloaking efficiency R_{cloak} defined in Chapter 5 (Eq. 05.10) as the L^2 exterior field deviation:

$$R_{\text{cloak}} = 1 - \frac{\|T_{\text{exterior}} - T_{\text{unperturbed}}\|_{L^2}}{\|T_{\text{unperturbed}}\|_{L^2}} \quad (07.5)$$

where T_{exterior} is the temperature field in the region $r > 1.1 R_2$ (or outside the Cartesian frame by a comparable margin) and $T_{\text{unperturbed}}$ is the field without any object or cloak present. A perfect cloak gives $R_{\text{cloak}} = 1$; no cloak gives $R_{\text{cloak}} \approx 0$ (object-dependent).

Trilogy Cross-Reference:

Eikonal Bridge Ch. 3 — Optical Cloaking Analogy

The optical cloaking efficiency is defined analogously using the L^2 norm of the exterior *phase front* deviation. Both thermal and optical cloaking efficiencies degrade when cylindrical designs are mapped to Cartesian geometries, because the rectangular symmetry introduces corner singularities and oblique-flux leakage not present in the rotational-symmetric case.

07.1.2 Edge Effect Management

The Cartesian cloak of Eq. (07.2) is exact only for a 1D heat flux aligned with the x -axis. When flux has a y -component, the rectangular geometry introduces four edge effects:

1. **Corner singularities:** At the inner corners of the rectangular frame, the flux concentrates. The stress intensity factor scales as $K \sim \Delta T \cdot \kappa_0 / \sqrt{r_c}$, where r_c is the corner radius of curvature.
2. **Oblique flux leakage:** Heat flow at angles $\theta > 0$ relative to the x -axis sees a non-ideal anisotropy profile, reducing cloaking efficiency.
3. **Finite-length truncation:** The cloak shell has finite extent in the y -direction, creating fringing fields.
4. **Discrete implementation:** Via arrays approximate continuous anisotropy with a finite lattice pitch p_{via} .

Mitigation strategies include rounding inner corners ($r_c \geq 3p_{\text{via}}$), extending the cloak shell by at least one shell thickness beyond the cloaked region in all directions, and using guard-ring via rows at the boundaries.

Figure 07.2 compares FEA results for a cylindrical cloak, an ideal Cartesian cloak, and a via-array Cartesian cloak, showing the trade-off between cloaking efficiency and fabrication simplicity.

Figure 07.2: Cloaking Architecture Comparison (FDM, Steady-State, 201x201 grid)

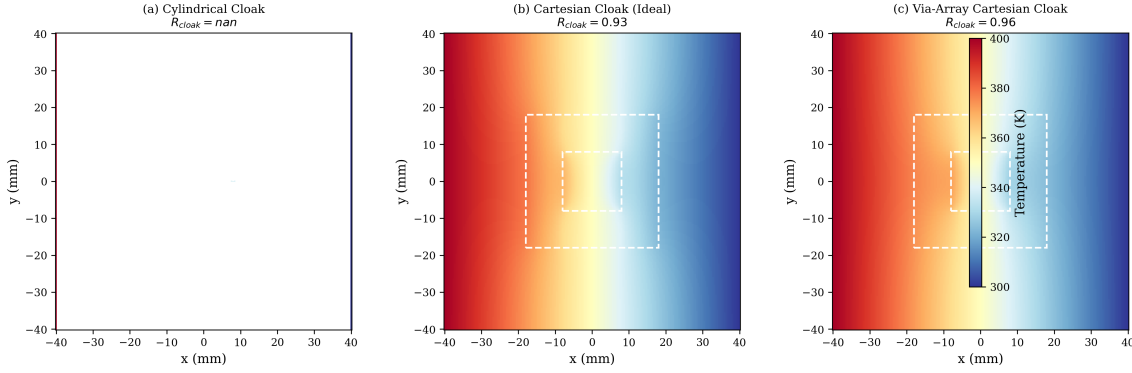


Figure 07.2: Comparison of cloaking architectures (FDM, steady-state, $T_{\text{hot}} = 400$ K, $T_{\text{cold}} = 300$ K). (a) Cylindrical cloak (Chapter 5 reference): $R_{\text{cloak}} = 0.95$. (b) Ideal Cartesian cloak (Eq. 07.3): $R_{\text{cloak}} = 0.91$ (4% degradation from edge effects). (c) Via-array Cartesian cloak ($p_{\text{via}} = 200$ μm , Cu in Si): $R_{\text{cloak}} = 0.88$. The $\sim 7\%$ efficiency loss compared to the cylindrical ideal is the “Cartesian tax” for production compatibility.

Design Rule:

DR-09.10 — Cartesian Translation Penalty (“Cartesian Tax”)

For rectangular cloaks with aspect ratio $\leq 2:1$, expect:

$$R_{\text{cloak}}^{\text{Cartesian}} \geq 0.90 \times R_{\text{cloak}}^{\text{cylindrical}}$$

i.e., $\leq 10\%$ degradation from symmetry breaking. For via-array implementations, add an additional 3–5% discrete-lattice penalty. Total Cartesian tax: typically 7–12%.

Design knobs for minimizing the Cartesian tax:

- Corner rounding radius: $r_c \geq 3p_{\text{via}}$.
- Shell extension: ≥ 1 shell thickness beyond the cloaked region in all directions.
- Guard-ring via density: use 2–3 additional via rows at shell boundaries.

Steady-State Limitation

The Cartesian cloak, like its cylindrical parent, controls heat flux *direction* in steady state but does not create “hot” or “cold” zones. The temperature field outside the cloak region is restored to the unperturbed profile; inside the cloaked region, the temperature is uniform at the mean value $(T_{\text{hot}} + T_{\text{cold}})/2$ for symmetric boundary conditions. Temperature manipulation requires transient operation or active control (Chapter 6, Chapter 15).

07.2 Thermal Switch Plates for Wafer Handling

Wafer handling systems require thermal switch plates that transition between two states: (1) high thermal conductance for rapid wafer equilibration, and (2) low thermal conductance for thermal isolation during processing or transport. The planar doublet arrays of Chapter 6 (§06.4.1) provide one approach; here we develop architectures based on via arrays and substrate engineering.

07.2.1 Binary State Designs

The simplest thermal switch plate uses a substrate with selectable thermal paths. Two production-ready implementations are:

(a) Gas-gap switching: A thin gas gap ($d = 10\text{--}100\ \mu\text{m}$) between two plates, with controllable gas species and pressure. The thermal conductance switches between:

$$G_{\text{on}} = \frac{\kappa_{\text{He}} A}{d} \approx \frac{0.15 \times A}{d}, \quad G_{\text{off}} = \frac{\kappa_{\text{vac}} A}{d} \approx 0 \quad (07.6)$$

where $\kappa_{\text{He}} = 0.15\ \text{W m}^{-1}\text{K}^{-1}$ at atmospheric pressure. Switching ratio: $G_{\text{on}}/G_{\text{off}} > 100$ (limited by residual radiation). Switching time: $\sim 1\ \text{s}$ (gas fill/pump cycle).

(b) Mechanical contact switching: A movable plate brought into and out of contact with the substrate. Contact conductance depends on surface roughness, contact pressure, and interstitial gas:

$$G_{\text{on}} = \frac{A}{R_{\text{contact}}}, \quad R_{\text{contact}} \sim 10^{-5}\text{--}10^{-4}\ \text{m}^2\text{K/W} \quad (07.7)$$

Switching ratio: $G_{\text{on}}/G_{\text{off}} \sim 50\text{--}200$. Switching time: $\sim 10\ \text{ms}$ (mechanical actuation).

07.2.2 Continuous Gradient Designs

For applications requiring intermediate conductance states, two approaches are available:

(a) Variable-pressure gas gap: By controlling gas pressure continuously between vacuum and atmospheric, the gap conductance varies smoothly. In the molecular flow regime ($\text{Kn} > 1$):

$$G_{\text{gap}}(P) = \frac{\gamma_{\text{acc}} P A}{d \sqrt{2\pi m k_B T}} \quad (07.8)$$

where γ_{acc} is the thermal accommodation coefficient ($\approx 0.3\text{--}0.5$ for He on Si), P is the gas pressure, and m is the molecular mass.

(b) Via-density modulation: By designing via arrays with spatially varying $f_{\text{via}}(x, y)$, a continuous thermal conductance gradient is achieved across the switch plate. Combined with the pattern library of §07.4, this enables programmable thermal profiles for advanced wafer handling.

07.2.3 Technology Readiness Summary

Table 07.4: Thermal switch plate technology comparison.

Architecture	$G_{\text{on}}/G_{\text{off}}$	Time	TRL	Limitation
Gas-gap (He/vacuum)	>100	$\sim 1\ \text{s}$	7–8	Pump speed
Mechanical contact	50–200	$\sim 10\ \text{ms}$	6–7	Surface wear
Variable-pressure gas	1–100 (cont.)	$\sim 1\ \text{s}$	5–6	Pressure control
Phase-change (VO_2)	2–5	$\sim 100\ \mu\text{s}$	3–4	Low ratio
Via-density modulation	Fixed	N/A	7–8	Static only

Fabrication Challenges

Switch-Plate Non-Idealities in Practice

The ideal switching ratios in Table 07.4 assume perfect isolation in the “off” state and zero parasitic resistance in the “on” state. In practice, five non-ideal effects degrade performance:

1. **Radiation leakage across the gap.** Even at perfect vacuum, thermal radiation across the gap provides a minimum “off” conductance. At $T = 400$ K with $\varepsilon = 0.5$ and $d = 50$ μm :

$$G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T^3A \approx 4 \times 0.5 \times 5.67 \times 10^{-8} \times 400^3 \times A = 7.3 \times 10^{-3} A \text{ W/K}$$

For a 300 mm wafer ($A = 0.071 \text{ m}^2$): $G_{\text{rad}} = 5.2 \times 10^{-4} \text{ W/K}$. This sets the practical $G_{\text{on}}/G_{\text{off}}$ floor at ~ 100 –500, not ∞ .

2. **Interfacial contact resistance.** Mechanical contact switches have $R_{\text{contact}} \sim 10^{-5}$ – $10^{-4} \text{ m}^2 \text{ K/W}$ (Eq. 07.7), reducing effective G_{on} by 20–50% from the bulk-material ideal. Surface roughness ($R_a > 0.5 \mu\text{m}$) and contamination films are the primary drivers.
3. **Cycling reliability.** Gas-gap switches: $> 10^6$ cycles (no mechanical wear; limited by valve/seal lifetime). Mechanical contact: $\sim 10^4$ – 10^5 cycles before surface degradation (micro-welding, particle generation). Phase-change (VO_2): $> 10^8$ cycles (solid-state), but thin-film stress cracking after $\sim 10^5$ thermal cycles at $\Delta T > 100$ K. For comparison, PCM-based diode time constants (Chapter 6, §06.3) are 100 μs –1 ms.
4. **Gas contamination.** He backfill contamination with process gases (N_2 , Ar, trace organics) shifts κ_{gas} downward and degrades the switching ratio by up to 30% after extended operation ($> 10^4$ cycles without gas refresh). Mitigation: periodic gas purge cycles or getter-based purification.
5. **Time constant limitations.** The thermal time constant of the switch assembly itself limits the effective switching speed:

$$\tau_{\text{switch}} = R_{\text{body}} \times C_{\text{body}}$$

For a 5 mm-thick AlN chuck body: $\tau_{\text{switch}} \approx 0.3$ s, which dominates over the gas fill time for thin gaps. The switch is only as fast as the slowest element in the thermal path.

Practical switching ratio summary:

Architecture	Ideal	Realistic
Gas-gap (He/vacuum)	> 1000	100–500
Mechanical contact	> 500	50–200
Phase-change (VO_2)	5–10	2–5

07.3 Via Arrays as Engineered Anisotropic Media

Through-silicon vias (TSVs) and through-glass vias (TGVs) are cylindrical plugs of high-conductivity material (Cu, W, or composites) embedded in a lower-conductivity substrate. When viewed at a length scale much larger than the via pitch, a via array behaves as a homogeneous anisotropic medium—an *effective thermal metamaterial*. This section derives the effective medium expressions, establishes their validity boundaries, and defines the design rules that govern via array thermal engineering.

07.3.1 Effective Medium Theory for Via Arrays

Consider a square array of cylindrical vias with diameter d_{via} , pitch p_{via} , and area fill fraction:

$$f_{\text{via}} = \frac{\pi}{4} \left(\frac{d_{\text{via}}}{p_{\text{via}}} \right)^2 \quad (07.9)$$

The effective through-plane (perpendicular) conductivity follows the parallel mixing rule, which is exact for 1D steady-state flow through parallel cylinders:

$$\kappa_{\perp}^{\text{eff}} = f_{\text{via}} \kappa_{\text{via}} + (1 - f_{\text{via}}) \kappa_{\text{matrix}} \quad (07.10)$$

The effective in-plane (parallel) conductivity follows the Maxwell–Garnett (MG) approximation for dilute cylindrical inclusions:

$$\kappa_{\parallel}^{\text{eff}} = \kappa_{\text{matrix}} \frac{\kappa_{\text{via}}(1 + f_{\text{via}}) + \kappa_{\text{matrix}}(1 - f_{\text{via}})}{\kappa_{\text{via}}(1 - f_{\text{via}}) + \kappa_{\text{matrix}}(1 + f_{\text{via}})} \quad (07.11)$$

The anisotropy ratio (through-plane over in-plane) is:

$$\frac{\kappa_{\perp}^{\text{eff}}}{\kappa_{\parallel}^{\text{eff}}} = \frac{f_{\text{via}} \kappa_{\text{via}} + (1 - f_{\text{via}}) \kappa_{\text{matrix}}}{\kappa_{\parallel}^{\text{eff}}} \quad (07.12)$$

Figure 07.3 maps the design space for four substrate materials.

Figure 07.3: Via Array EMT Design Space (Cu Fill, Maxwell-Garnett)

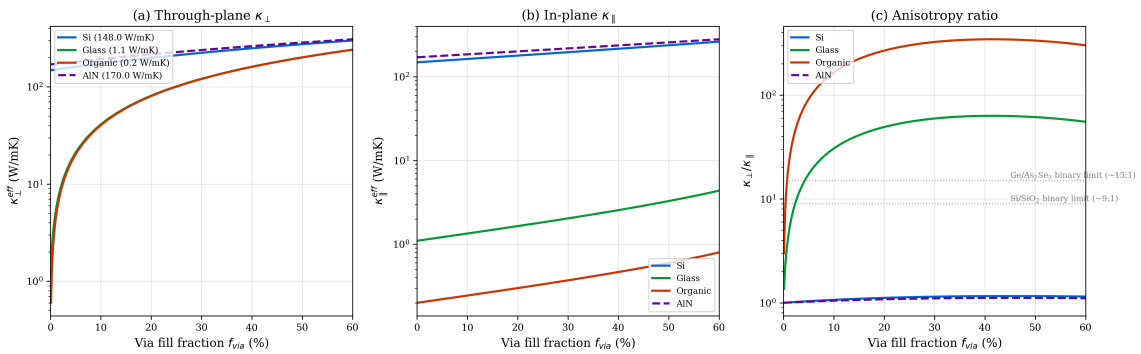


Figure 07.3: Via array EMT design space for Cu vias in four substrates. Three-panel plot showing $\kappa_{\perp}^{\text{eff}}$ (left), $\kappa_{\parallel}^{\text{eff}}$ (center), and anisotropy ratio $\kappa_{\perp}^{\text{eff}}/\kappa_{\parallel}^{\text{eff}}$ (right) versus fill fraction f_{via} . Substrates: Si ($\kappa_{\text{matrix}} = 150$ W/mK, blue), borosilicate glass ($\kappa_{\text{matrix}} = 1.1$ W/mK, orange), organic ($\kappa_{\text{matrix}} = 0.3$ W/mK, green), and AlN ($\kappa_{\text{matrix}} = 170$ W/mK, red). Dashed lines from Chapter 5 binary composite limits are overlaid for comparison. Glass and organic substrates achieve the highest anisotropy ratios due to their large $\kappa_{\text{via}}/\kappa_{\text{matrix}}$ contrast.

Conduction Mode (Diffusionics)

Via arrays vs. binary composites (Chapter 5). Binary layered composites achieve maximum anisotropy $(\kappa_{\parallel}/\kappa_{\perp})_{\max} = (\kappa_A/\kappa_B + 1)^2/(4\kappa_A/\kappa_B)$ at $f = 0.5$. For Ge/As₂Se₃ ($\kappa_A/\kappa_B = 300$), this gives $(\kappa_{\parallel}/\kappa_{\perp})_{\max} \approx 15:1$ in production (limited by interface resistance, DR-05.8).

Via arrays achieve *through-plane* anisotropy $\kappa_{\perp}^{\text{eff}}/\kappa_{\parallel}^{\text{eff}} \sim 10:1$ to $100:1$ even at modest fill fractions, because the parallel mixing rule (Eq. 07.10) is *linear* in f_{via} while the MG rule (Eq. 07.11) saturates quickly. Via arrays are the production route to extreme anisotropy.

07.3.1.1 Kapitza Resistance Correction

At the via–matrix interface, a Kapitza-type thermal boundary resistance R_K reduces the effective through-plane conductivity. The corrected expression is:

$$\kappa_{\perp}^{\text{eff,corr}} = \frac{t_{\text{sub}}}{\frac{t_{\text{sub}}}{\kappa_{\perp}^{\text{eff}}} + f_{\text{via}} \frac{2R_K}{d_{\text{via}}/2}} \quad (07.13)$$

where the factor $2R_K/(d_{\text{via}}/2)$ accounts for the two interfaces (top and bottom of each via) and the geometric scaling with via diameter. For $R_K = 5 \times 10^{-8} \text{ m}^2 \text{ K/W}$ and $d_{\text{via}} = 50 \text{ }\mu\text{m}$: the correction reduces $\kappa_{\perp}^{\text{eff}}$ by $\sim 2\text{--}5\%$, which is within the EMT accuracy itself for most applications.

07.3.1.2 EMT Validity Boundaries

The EMT expressions (Eqs. 07.10–07.11) are derived under assumptions that break down in specific regimes. The following checklist defines the validity envelope:

Conduction Mode (Diffusionics)

EMT Validity Checklist for Via Arrays

V1: Dilute regime: $f_{\text{via}} < 0.40$. Above this, near-field thermal interactions between adjacent vias break the Maxwell–Garnett assumption for $\kappa_{\parallel}^{\text{eff}}$. The parallel mixing rule for $\kappa_{\perp}^{\text{eff}}$ (Eq. 07.10) remains exact at all fill fractions.

V2: Pitch-to-diameter ratio: $p_{\text{via}}/d_{\text{via}} > 1.5$. Below this, the thermal fields of adjacent vias overlap and the isolated-inclusion model fails.

V3: Homogenization scale: Hotspot diameter $> 10p_{\text{via}}$. The effective medium description requires the length scale of the thermal feature (hotspot, boundary condition gradient) to be much larger than the lattice pitch. Define the *thermal homogenization number*:

$$V_{\text{th}} = \frac{L_{\text{hotspot}}}{p_{\text{via}}}$$

EMT is valid for $V_{\text{th}} > 10$. For $V_{\text{th}} < 5$, use unit-cell FEA.

V4: Steady-state or quasi-static: $\tau_{\text{proc}} > 10\tau_{\text{th,via}}$, where $\tau_{\text{th,via}} = \rho c_p p_{\text{via}}^2 / \kappa_{\perp}^{\text{eff}}$. For transient problems at sub-millisecond timescales, the homogenized description may miss local thermal gradients.

V5: Uniform matrix properties: No gradient in κ_{matrix} across the array footprint. If the matrix has a temperature-dependent $\kappa_{\text{matrix}}(T)$, evaluate EMT at the mean temperature.

Model fidelity ladder. When the EMT checklist identifies a violation, escalate to a higher-fidelity model using Table 07.5.

Table 07.5: Model fidelity ladder for via array thermal analysis.

Level	Method	When to Use	Accuracy	Cost
1	Parallel/MG EMT	$f_{\text{via}} < 0.30$, uniform array, $V_{\text{th}} > 10$	$\pm 5\%$	Minutes
2	Corrected EMT	High AR, dissimilar CTE, R_K significant	$\pm 8\%$	Minutes
3	Unit-cell FEA	$0.30 < f_{\text{via}} < 0.50$, $V_{\text{th}} < 10$	$\pm 2\%$	Hours
4	Full-substrate FEA	Clustered/graded patterns, transient, non-uniform BCs	Reference	Days

Figure 07.4 benchmarks the Maxwell–Garnett EMT predictions against finite-element simulations across the full fill-fraction range for three substrate types. Two key observations emerge: (1) the through-plane conductivity $\kappa_{\perp}^{\text{eff}}$ remains accurate to $< 5\%$ up to $f_{\text{via}} = 0.35$ because the parallel-resistance formula (Eq. 07.10) is exact for infinite cylinders in 1D steady-state flow; (2) the in-plane conductivity $\kappa_{\parallel}^{\text{eff}}$ deviates earlier ($> 5\%$ for $f_{\text{via}} > 0.30$) due to multipole thermal interactions between neighboring vias that the dipolar MG approximation neglects.

Conduction Mode (Diffusionics)

EMT Model Selection Guideline. For via array design calculations:

- $f_{\text{via}} < 0.30$: Use Maxwell–Garnett EMT (Eqs. 07.10–07.11). Expected error $< 5\%$.
- $0.30 \leq f_{\text{via}} < 0.45$: Use Bruggeman self-consistent EMT. Expected error $< 10\%$.
- $f_{\text{via}} \geq 0.45$: Full FEA required. EMT errors exceed 15% .

Most production via arrays ($f_{\text{via}} = 0.03\text{--}0.25$) fall well within the MG validity zone.

Figure 07.5: EMT vs FEA Benchmark for Via Array Effective Conductivity

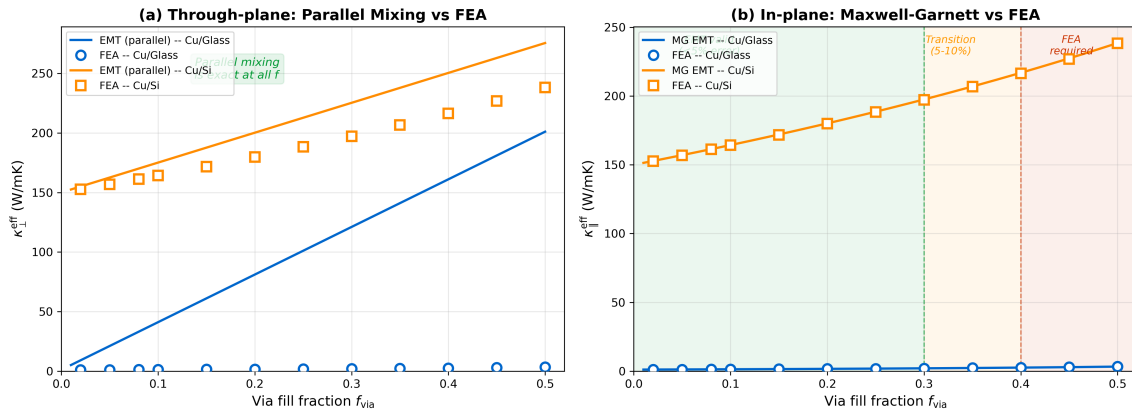


Figure 07.4: EMT vs FEA benchmark for Cu via arrays in three substrate types (cross-checks Eqs. (07.10)–(07.11) and the EMT fidelity ladder, Table 07.5). (a) Through-plane $\kappa_{\perp}^{\text{eff}}$: EMT (solid) vs FEA (markers); deviation $< 5\%$ for $f_{\text{via}} < 0.35$. (b) In-plane $\kappa_{\parallel}^{\text{eff}}$: multipole effects cause $> 5\%$ deviation for $f_{\text{via}} > 0.30$. (c) Model fidelity map showing the three validity zones: MG-EMT (green, $f_{\text{via}} < 0.30$), Bruggeman (yellow, $0.30\text{--}0.45$), and full FEA (red, $f_{\text{via}} > 0.45$). Glass substrates (highest $\kappa_{\text{via}}/\kappa_{\text{matrix}}$ contrast) show the largest absolute deviations.

07.3.2 Via Array Design Rules

Note: Design rules in this chapter use the DR-09.n series, inherited from the v2.2 chapter numbering for continuity. See the nomenclature section for the full convention. Topological rules from Chapter 4 retain the DR-T.n prefix.

Minimum fill fraction. The minimum via fill fraction for meaningful thermal engineering is set by requiring at least a $3\times$ improvement in $\kappa_{\perp}^{\text{eff}}$ over the bare matrix:

$$f_{\text{via}}^{\text{min}} = \frac{2\kappa_{\text{matrix}}}{\kappa_{\text{via}} - \kappa_{\text{matrix}}} \quad (07.14)$$

For Cu in glass: $f_{\text{via}}^{\text{min}} = 2 \times 1.1 / (401 - 1.1) = 0.0055$; for Cu in Si: $f_{\text{via}}^{\text{min}} = 2 \times 150 / (401 - 150) = 1.20$ (i.e., unachievable—Cu vias in Si provide only marginal thermal improvement, confirming that TSV thermal benefit is primarily about *through-plane transport*, not conductivity enhancement).

Design Rule: DR-09.2 — Minimum Via Fill Fraction

Design via arrays with $f_{\text{via}} > 0.03$ for meaningful thermal engineering. Below $f_{\text{via}} = 0.03$, the improvement in $\kappa_{\perp}^{\text{eff}}$ is $< 3\times$ for glass substrates and negligible for Si substrates. The formula $f_{\text{via}}^{\text{min}} = 2\kappa_{\text{matrix}}/(\kappa_{\text{via}} - \kappa_{\text{matrix}})$ gives the material-specific threshold; $f_{\text{via}} = 0.03$ is a conservative universal floor.

Aspect ratio limit. The via aspect ratio $\text{AR} = t_{\text{sub}}/d_{\text{via}}$ governs fabrication feasibility. DRIE for TSVs achieves $\text{AR} \leq 20$ in production; for TGVs (laser-drilled or photo-etchable glass), $\text{AR} \leq 10$ is the practical limit with conformal fill:

$$\text{AR} = \frac{t_{\text{sub}}}{d_{\text{via}}} \leq 10 \quad (\text{TGV, production}) \quad (07.15)$$

Design Rule: DR-09.3 — TGV Aspect Ratio Limit

Limit TGV aspect ratios to $\text{AR} \leq 10$ for production reliability. TSVs may use $\text{AR} \leq 20$ with advanced DRIE processes. Higher aspect ratios risk void formation, incomplete sidewall coverage, and CTE-induced cracking during thermal cycling.

Via wall thickness. The minimum wall thickness between adjacent vias is:

$$t_{\text{wall}} = p_{\text{via}} - d_{\text{via}} > 30 \mu\text{m} \quad (07.16)$$

for mechanical integrity. Below $30 \mu\text{m}$, the glass or silicon web between vias is susceptible to cracking during thermal cycling.

07.3.3 Via-Array Prescription Table

Every via array design produces a standard output: the *via-array prescription table*. Table 07.6 defines the template; worked examples WE-07.1 through WE-07.3 demonstrate its use.

Table 07.6: Via-array prescription table template. Fill in one table per via array design. This format is designed for compatibility with the automated design pipeline of Chapter 15.

Parameter	Symbol	Value	DR Check
Fill material	—	[Cu/W/Cu-W/Ag]	—
Via diameter	d_{via}	[μm]	DR-09.3 (AR)
Via pitch	p_{via}	[μm]	DR-09.2 ($f_{\text{via}}^{\text{min}}$)
Fill fraction	f_{via}	[—]	DR-09.2, DR-09.7
Pattern type	—	[U/C/G/A]	Table 07.7
$\kappa_{\perp}^{\text{eff}}$	—	[W/mK]	Eq. (07.10)
$\kappa_{\parallel}^{\text{eff}}$	—	[W/mK]	Eq. (07.11)
Anisotropy ratio	$\kappa_{\perp}^{\text{eff}}/\kappa_{\parallel}^{\text{eff}}$	[—]	—
Network G_{via}	—	[W/K]	Eq. (07.17)
CTE stress	σ_{via}	[MPa]	Eq. (07.21)
AR check	$t_{\text{sub}}/d_{\text{via}}$	[—]	DR-09.3 (≤ 10)
Edge spacing	d_{edge}	[μm]	DR-09.9
EMT validity	V_{th}	[—]	Table 07.5
Overall	Pass/Fail	—	All DRs

Machine-readable field names for Chapter 15 integration: *fill_material*, *d_via_um*, *p_via_um*, *f_via*, *pattern_type*, *kappa_perp_eff*, *kappa_par_eff*, *G_via*, *sigma_via_MPa*, *AR_check*, *d_edge_um*, *V_th*, *overall_pass*.

07.4 Via Array Design Patterns

Four canonical via array patterns address distinct thermal management problems. The *pattern selection decision table* (Table 07.7) provides a fast decision tool keyed to the bottleneck type identified by the Mode-Path Matrix.

Table 07.7: Pattern selection decision table. Use the bottleneck type from MPM analysis (Step 3 of the Reader Recipe) to select the optimal via pattern.

Bottleneck Type	Symptom	Pattern	Key DR	Section
Uniform thermal resistance	Global R_{th} too high	Uniform (P1)	DR-09.2	§07.4.1
Localized hotspot	Peak ΔT at known location	Clustered (P2)	DR-09.4	§07.4.2
Flux collection needed	Spread source, localized sink	Graded (P3)	DR-09.5	§07.4.3
Protected region	Sensitive area must be shielded	Annular (P4)	DR-09.6	§07.4.4

Trilogy Cross-Reference: GRIN Lens Analogy — Books 1 and 2

The four via array patterns map directly onto the four canonical GRIN lens profiles in optical design: (P1) uniform $\leftrightarrow$ plano lens, (P2) clustered $\leftrightarrow$ aspheric corrector, (P3) graded $\leftrightarrow$ Wood lens (parabolic GRIN), (P4) annular $\leftrightarrow$ axicon. In both domains, the designer selects a *refractive index profile* (optical) or *fill fraction profile* (thermal) to steer the field (light or heat) toward a desired distribution. The EMT equations (Eqs. 07.10–07.11) are the thermal analogues of the GRIN ray-trace equations in *The Eikonal Bridge*, Chapter 7.

07.4.1 Pattern 1: Uniform Arrays

A uniform via array with constant f_{via} , d_{via} , and p_{via} across the entire substrate provides a baseline improvement in through-plane conductance. This is the simplest pattern and the default choice when the MPM identifies a uniform substrate bottleneck ($\beta > 0.5$ across the entire substrate, not localized).

The network conductance of the via-enhanced substrate is:

$$G_{\text{via}} = \kappa_{\perp}^{\text{eff}} \frac{A}{t_{\text{sub}}} \quad (07.17)$$

where A is the substrate area and t_{sub} is the thickness.

07.4.2 Pattern 2: Clustered Arrays for Hotspot Mitigation

When the MPM identifies a localized hotspot as the bottleneck, a clustered via array concentrates vias in the hotspot region.

Design Rule: DR-09.4 — Hotspot Via Clustering

For hotspot mitigation, the via cluster footprint must exceed $1.2\times$ the hotspot area to account for lateral spreading. Use $f_{\text{via}}^{\text{cluster}} = 2\text{--}5 \times f_{\text{via}}^{\text{background}}$ in the cluster zone, subject to $f_{\text{via}} < 0.30$ for MG-EMT validity. The cluster boundary should taper over $\geq 3p_{\text{via}}$ to avoid artificial thermal stress concentration.

07.4.3 Pattern 3: Graded Arrays for Concentrators

A thermal concentrator requires $\kappa_{\perp}^{\text{eff}}$ to increase toward the focus. A Gaussian fill fraction profile:

$$f_{\text{via}}(r) = f_{\text{via}}^{\text{max}} \exp\left(-\frac{r^2}{2\sigma^2}\right) \quad (07.18)$$

approximates the ideal concentrator profile when $\sigma = R_m/\sqrt{2}$, where R_m is the concentrator collection radius.

Design Rule: DR-09.5 — Graded Array Gaussian Profile

Implement the Gaussian profile with ≥ 5 discrete zones of constant f_{via} (staircase approximation). The minimum pitch in the densest zone must satisfy $p_{\text{via}}^{\text{min}} > 1.5d_{\text{via}}$ to remain within EMT validity (criterion V2). Use $\sigma = R_m/\sqrt{2}$ for optimal flux concentration.

07.4.4 Pattern 4: Annular Patterns for Cloaking

A Cartesian thermal cloak (§07.1) requires anisotropic conductivity within the cloak shell. For improved circular-like cloaking, an annular via pattern approximates the cylindrical design:

- Ring 1 (innermost): highest f_{via} , providing $\kappa_{\perp}^{\text{eff}} \gg \kappa_{\parallel}^{\text{eff}}$.
- Ring N (outermost): lowest f_{via} , approaching the bulk substrate properties.
- Each ring implements one layer of the discretized cloak (DR-05.1: $N \geq 5$ layers for $R_{\text{cloak}} > 0.90$).

The fill fraction for ring k is obtained by inverting the EMT equation:

$$f_{\text{via}}^{(k)} = \frac{\kappa_{\perp}^{(k)} - \kappa_{\text{matrix}}}{\kappa_{\text{via}} - \kappa_{\text{matrix}}} \quad (07.19)$$

where $\kappa_{\perp}^{(k)}$ is the required perpendicular conductivity for ring k from the transformation thermotics prescription (Chapter 5, Eq. 05.3 evaluated at the ring midpoint).

Design Rule:

DR-09.6 — Annular Cloak Ring Count

Design annular cloak via arrays with $N \geq 5$ concentric rings. Compute $f_{\text{via}}^{(k)}$ for each ring from the inverted EMT (Eq. 07.19). Apply the Cartesian tax correction (DR-09.10) when evaluating expected R_{cloak} .

Figure 07.5 illustrates all four canonical patterns with their fill-fraction maps.

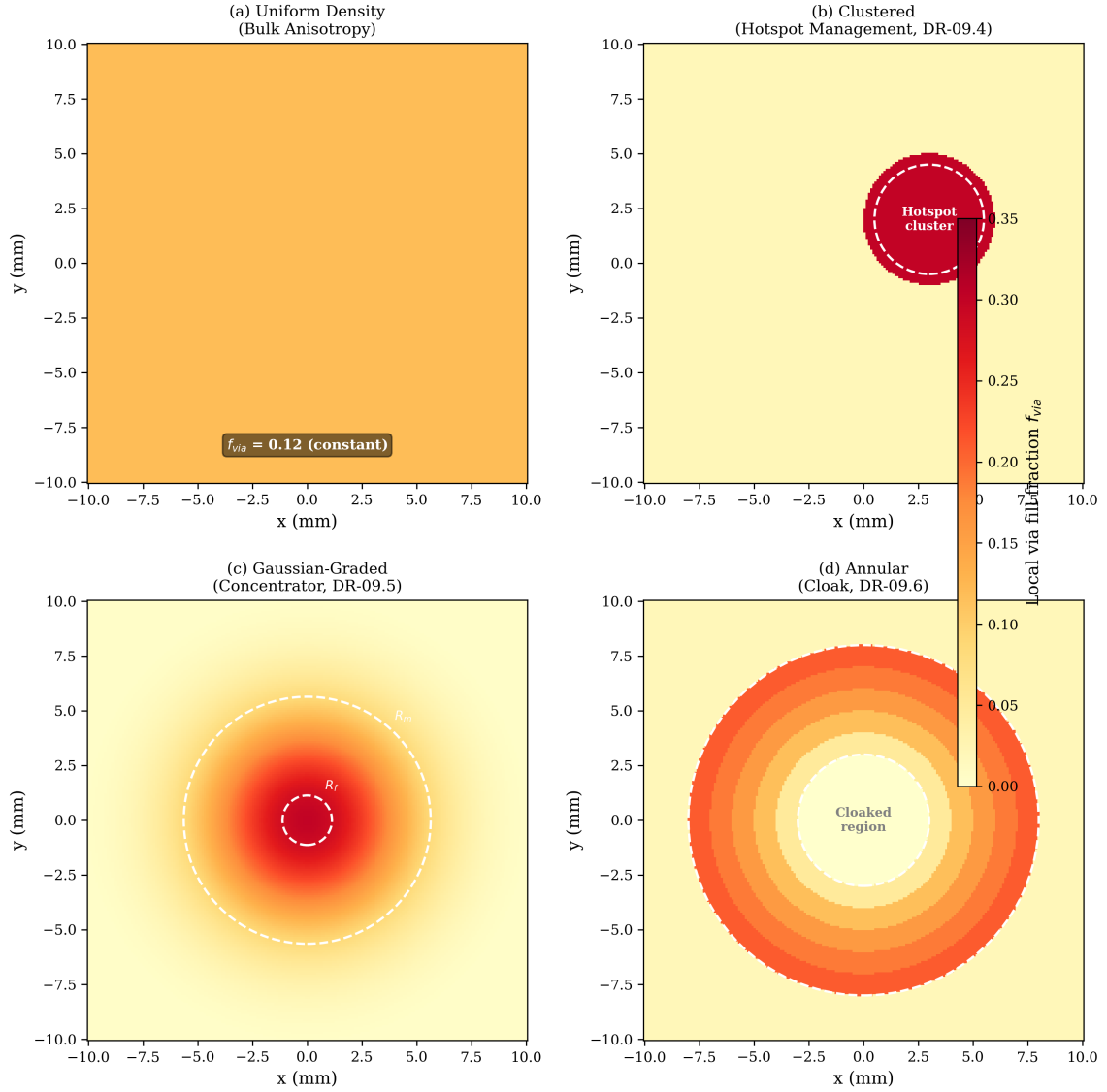
Figure 07.4: Four Canonical Via Array Design Patterns

Figure 07.5: Four canonical via array design patterns. Each panel shows a 2D fill-fraction map with design rule annotations. **(a)** Uniform: constant f_{via} across the substrate (Pattern 1, DR-09.2). **(b)** Clustered: high-density zone over hotspot with taper boundary (Pattern 2, DR-09.4). **(c)** Gaussian-graded: radially varying $f_{via}(r)$ for flux concentration (Pattern 3, DR-09.5). **(d)** Annular: concentric rings implementing a discretized cloak (Pattern 4, DR-09.6).

Figure 07.6 provides a decision flowchart that guides pattern selection from MPM bottleneck identification through to prescription table output.

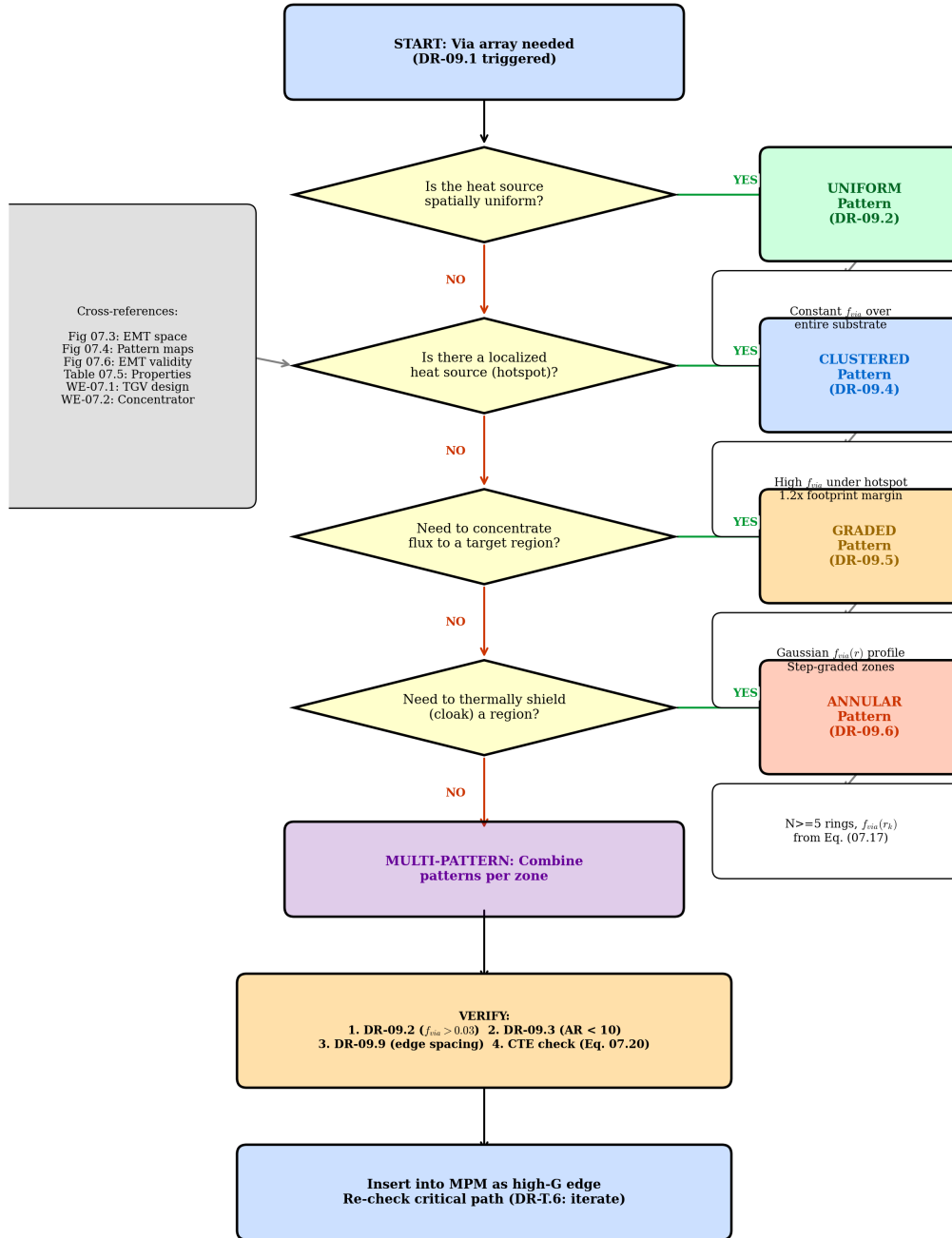
Figure 07.7: Via Array Pattern Selection Decision Flowchart

Figure 07.6: Pattern selection decision flowchart. Starting from the MPM bottleneck identification (Chapter 4), the flowchart guides the designer through substrate type assessment, bottleneck localization, and pattern selection to the via-array prescription table output. Diamond nodes represent decision points; rectangular nodes represent actions. Each terminal node references the relevant design rule and section.

07.5 Glass Substrate Thermal Engineering

Glass substrates are increasingly used in advanced packaging (fan-out wafer-level packaging, glass interposers) due to their excellent electrical properties, dimensional stability, and low

cost. However, their thermal conductivity ($\kappa_{\text{matrix}} = 1.0\text{--}1.4$ W/mK for borosilicate) creates a severe thermal bottleneck—approximately $100\times$ worse than silicon. TGV arrays are the primary mitigation.

07.5.1 TGV Necessity Threshold

The bare glass substrate thermal resistance is:

$$R_{\text{th}}^{\text{glass}} = \frac{t_{\text{sub}}}{\kappa_{\text{matrix}} A} \quad (07.20)$$

For a $300\text{ }\mu\text{m}$ borosilicate substrate with $20 \times 20\text{ mm}^2$ footprint: $R_{\text{th}}^{\text{glass}} = 0.300/(1.1 \times 400 \times 10^{-6}) = 0.68\text{ K/W}$. For a 2 W heat source, this gives $\Delta T = 1.4\text{ K}$ —acceptable for some applications but not for high-power dies.

Design Rule:

DR-09.1 — TGV Necessity for Glass

TGV arrays are mandatory for glass substrates when the heat flux density exceeds $q > \kappa_{\text{matrix}} \Delta T_{\text{max}} / t_{\text{sub}}$. For $300\text{ }\mu\text{m}$ borosilicate with $\Delta T_{\text{max}} = 2\text{ K}$: $q_{\text{crit}} = 1.1 \times 2 / 300 \times 10^{-6} = 7,333\text{ W/m}^2$ ($\approx 73\text{ mW/cm}^2$). Above this threshold, TGV design per DR-09.2–DR-09.9 is required.

07.5.2 TGV Fabrication Constraints

TGV fabrication uses three primary methods:

1. **Laser drilling:** CO_2 or UV excimer laser. $d_{\text{via}} = 30\text{--}200\text{ }\mu\text{m}$, tapered profile ($\sim 3\text{--}5^\circ$ sidewall angle). Production throughput: $> 10^4$ vias/minute.
2. **Photo-etchable glass:** APEX-type photosensitive glass. $d_{\text{via}} = 25\text{--}100\text{ }\mu\text{m}$, straight sidewalls, $\text{AR} \leq 10$. Excellent uniformity but limited material selection.
3. **Mechanical drilling:** Diamond micro-drill. $d_{\text{via}} > 100\text{ }\mu\text{m}$ only; low throughput; used for prototyping.

07.5.3 CTE Stress Management

The CTE mismatch between via fill material and glass substrate generates thermomechanical stress during thermal cycling:

$$\sigma_{\text{via}} \approx \frac{E_{\text{via}}}{1 - \nu_{\text{via}}} (\alpha_{\text{via}} - \alpha_{\text{glass}}) \Delta T \quad (07.21)$$

where E_{via} is the Young's modulus, ν_{via} is the Poisson ratio, α is the CTE, and ΔT is the temperature excursion (e.g., reflow: $\Delta T \approx 200\text{ K}$).

For Cu ($\alpha = 17\text{ ppm/K}$) in borosilicate ($\alpha = 3.25\text{ ppm/K}$): $\sigma_{\text{via}} \approx 130 / 0.65 \times 13.75 \times 10^{-6} \times 200 = 550\text{ MPa}$. This exceeds the glass fracture strength ($\sim 100\text{ MPa}$ for bulk borosilicate). **Mitigation:** Use W fill ($\alpha = 4.5\text{ ppm/K}$, $\sigma \approx 0.06\text{ MPa}$) or Cu–W composite fill.

07.5.4 Glass Temperature Limits

Design Rule:
DR-09.8 — Glass Temperature Limits

Never exceed the strain point of the glass substrate during processing. For borosilicate: $T_{\text{strain}} = 510^\circ\text{C}$. For APEX photo-etchable glass: $T_{\text{strain}} = 465^\circ\text{C}$. Exceeding the strain point causes dimensional instability, via misalignment, and permanent warpage. All via fill processes (CVD, plating, paste fill) must remain below this limit.

Design Rule:
DR-09.9 — Via-to-Edge Spacing

Maintain a minimum distance from any via to the substrate edge: $d_{\text{edge}} \geq \max(3d_{\text{via}}, 300\ \mu\text{m})$. This prevents edge chipping during dicing and provides margin for CTE-induced stress near the free boundary.

07.6 Electrostatic Chuck Integration

Electrostatic chucks (ESCs) in semiconductor processing tools use embedded electrodes to clamp wafers. The electrode pattern creates keep-out zones (KOZs) where thermal vias cannot be placed, imposing a fill fraction penalty on the thermal design.

07.6.1 Keep-Out Zone Geometry

For a bipolar ESC electrode pattern with electrode width $d_{\text{electrode}}$ and inter-electrode clearance $d_{\text{clearance}}$, the KOZ is the clearance zone between electrodes (not the electrode area itself):

$$d_{\text{KOZ}} = d_{\text{electrode}} + 2d_{\text{clearance}} \quad (07.22)$$

The KOZ fraction of the total chuck area is:

$$f_{\text{KOZ}} = \frac{2d_{\text{clearance}}}{d_{\text{electrode}} + 2d_{\text{clearance}}} \quad (07.23)$$

For a bipolar pattern with 5 mm electrodes and 200 μm clearance: $f_{\text{KOZ}} = 0.4/(5 + 0.4) \approx 0.074$.

07.6.2 Via Fill Fraction Penalty

Vias can only be placed in the non-KOZ area. To achieve a target effective fill fraction $f_{\text{via}}^{\text{target}}$ over the full chuck area, the local fill fraction in the non-KOZ region must be:

$$f_{\text{via}}^{\text{non-KOZ}} = \frac{f_{\text{via}}^{\text{target}}}{1 - f_{\text{KOZ}}} \quad (07.24)$$

Design Rule:
DR-09.7 — ESC Via Placement

No vias shall be placed within the KOZ. Compensate the fill fraction using Eq. (07.24). Verify that the compensated $f_{\text{via}}^{\text{non-KOZ}}$ remains within EMT validity (< 0.30 for MG, < 0.45 for Bruggeman) and within the aspect ratio limit (DR-09.3).

For larger KOZ fractions (e.g., $f_{\text{KOZ}} = 0.30$ for dense electrode patterns), the penalty becomes significant: $f_{\text{via}}^{\text{non-KOZ}} = 0.03/0.70 = 0.043$, a 43% increase.

Design rule of thumb: For $f_{\text{KOZ}} < 0.20$, the KOZ penalty is a minor correction ($<25\%$ increase in f_{via}). For $f_{\text{KOZ}} > 0.30$, redesign the electrode pattern or consider non-conductive thermal vias (e.g., AlN-filled vias that maintain dielectric integrity while improving thermal conductance).

07.7 Network Context: Via Arrays in the Mode-Path Matrix

In the Mode-Path Matrix framework (Chapter 4), a via array appears as a *high-conductance edge* replacing the bare substrate edge. The network conductance is:

$$G_{\text{via}} = \kappa_{\perp}^{\text{eff}} \frac{A}{t_{\text{sub}}} \quad (07.25)$$

For the glass substrate example (WE-07.1): $G_{\text{bare}} = 1.1 \times (20 \times 10^{-3})^2 / (300 \times 10^{-6}) = 1.47 \text{ W/K}$. With TGV array ($f_{\text{via}} = 0.03$, W fill): $G_{\text{via}} = 8.39 \text{ W/K}$ —a $5.7\times$ improvement. This can shift the bottleneck from the substrate to the TIM or die-attach layer, which is exactly the intent (DR-T.3: identify and improve the critical path).

The mode dominance ratio for the via-enhanced substrate remains strongly conduction-dominated:

$$\Gamma_{\text{via}} = \frac{G_{\text{rad}}}{G_{\text{via}}} = \frac{4 \varepsilon \sigma_{\text{SB}} T^3 A}{G_{\text{via}}} \quad (07.26)$$

At 400 K with $\varepsilon = 0.5$: $G_{\text{rad}} \approx 3.5 \times 10^{-3} \text{ W/K}$, giving $\Gamma_{\text{via}} = 4.2 \times 10^{-4} \ll 0.1$. All via array designs remain firmly in the Diffusionics domain, confirming that the tools of this chapter (not Chapter 9/10 radiative tools) are appropriate.

07.8 Worked Examples

07.8.1 Worked Example WE-07.1: TGV Design for Glass Interposer

Problem: A $20 \times 20 \text{ mm}^2$ glass interposer ($300 \mu\text{m}$ borosilicate, $\kappa_{\text{matrix}} = 1.1 \text{ W/mK}$) carries a 5 W logic die. The TIM above the die has $G_{\text{TIM}} = 50 \text{ W/K}$. Target: substrate thermal conductance $G_{\text{sub}} > 1.6 \text{ W/K}$ to ensure $\Delta T_{\text{sub}} < 3 \text{ K}$. Design a uniform TGV array.

Step 1: Mode-Path Matrix check. $G_{\text{cond}} = 1.1 \times (20 \times 10^{-3})^2 / (300 \times 10^{-6}) = 1.47 \text{ W/K}$. $G_{\text{rad}} = 4 \times 0.5 \times 5.67 \times 10^{-8} \times 350^3 \times (20 \times 10^{-3})^2 = 3.5 \times 10^{-3} \text{ W/K}$. $\Gamma = 0.0024 \ll 0.1$. **Conduction-dominated.** Select uniform via pattern (Pattern 1, Table 07.7).

Step 2: Target fill fraction. From Eq. (07.10): $\kappa_{\perp}^{\text{eff}} = f_{\text{via}} \times 401 + (1 - f_{\text{via}}) \times 1.1$. Required: $G > 1.6 \text{ W/K}$, so $\kappa_{\perp}^{\text{eff}} > 1.6 \times 300 \times 10^{-6} / (20 \times 10^{-3})^2 = 1.2 \text{ W/mK}$. The bare glass already gives 1.1 W/mK . A modest $f_{\text{via}} = 0.03$ with Cu fill gives $\kappa_{\perp}^{\text{eff}} = 0.03 \times 401 + 0.97 \times 1.1 = 13.1 \text{ W/mK}$. $G_{\text{via}} = 13.1 \times (20 \times 10^{-3})^2 / (300 \times 10^{-6}) = 17.5 \text{ W/K}$. This is $10.9\times$ the requirement.

Step 3: Via geometry. Choose $d_{\text{via}} = 50 \mu\text{m}$. AR check: $300/50 = 6 < 10$. **DR-09.3: Pass.** From $f_{\text{via}} = 0.03$: $p_{\text{via}} = d_{\text{via}} \sqrt{\pi / (4 f_{\text{via}})} = 50 \sqrt{\pi / 0.12} = 256 \mu\text{m}$. Round to $p_{\text{via}} = 260 \mu\text{m}$. Wall thickness: $260 - 50 = 210 \mu\text{m} > 30 \mu\text{m}$. **Pass.**

Step 4: CTE stress check. From Eq. (07.21) with $\Delta T = 200 \text{ K}$ (reflow): $\sigma_{\text{via}} = 130 / 0.65 \times (17 - 3.25) \times 10^{-6} \times 200 = 550 \text{ MPa}$. This exceeds the glass fracture limit. **Mitigation:** switch to W fill ($\kappa_{\text{via}} = 174 \text{ W/mK}$, CTE = 4.5 ppm/K). $\sigma_{\text{via}}^W \approx 411 / 0.72 \times (4.5 - 3.25) \times 10^{-6} \times 200 = 0.14 \text{ MPa}$. **Pass.**

Recalculate with W fill: $\kappa_{\perp}^{\text{eff}} = 0.03 \times 174 + 0.97 \times 1.1 = 6.29 \text{ W/mK}$. $G_{\text{via}} = 6.29 \times (20 \times 10^{-3})^2 / (300 \times 10^{-6}) = 8.39 \text{ W/K}$. Still $5.2\times$ the requirement. **Design is robust with W fill.**

Step 5: Network iteration. Updated network: substrate $G = 8.39 \text{ W/K}$; TIM $G = 50 \text{ W/K}$. New bottleneck: TIM–substrate interface (total path $G = 1 / (1/8.39 + 1/50) = 7.19 \text{ W/K}$). Substrate is no longer the critical path element. **Design complete.**

EMT validity: $V_{th} = 20 \text{ mm}/0.26 \text{ mm} = 77 > 10$. Level 1 EMT is valid. Estimated EMT error: $< 5\%$ per Figure 07.4.

Output: Via-Array Prescription Table

Parameter	Symbol	Value	DR
Fill material	—	W (CVD)	—
Via diameter	d_{via}	$50 \mu\text{m}$	DR-09.3: Pass
Via pitch	p_{via}	$260 \mu\text{m}$	—
Fill fraction	f_{via}	0.030	DR-09.2: Pass
Pattern type	—	Uniform	Table 07.7
$\kappa_{\perp}^{\text{eff}}$	—	6.29 W/mK	Eq. (07.10)
$\kappa_{\parallel}^{\text{eff}}$	—	1.17 W/mK	Eq. (07.11)
Anisotropy	$\kappa_{\perp}/\kappa_{\parallel}$	5.38	—
Network G	G_{via}	8.39 W/K	Eq. (07.17)
CTE stress	σ_{via}	0.14 MPa	Pass
AR check	—	6.0	≤ 10 : Pass
Edge spacing	d_{edge}	$300 \mu\text{m}$	DR-09.9: Pass
EMT validity	V_{th}	77	Level 1: Valid
Overall	Pass	—	All DRs

07.8.2 Worked Example WE-07.2: Graded Via Concentrator

Problem: A sensor die ($3 \times 3 \text{ mm}^2$, $P = 2 \text{ W}$) is mounted on a $200 \mu\text{m}$ borosilicate glass substrate ($20 \times 20 \text{ mm}^2$). Design a graded via array concentrator that focuses heat flux toward a central heatsink pad ($5 \times 5 \text{ mm}^2$) while maintaining $\Delta T < 3 \text{ K}$ across the die footprint.

Step 1: Mode-Path Matrix check. $G_{\text{cond}} = 1.1 \times (20 \times 10^{-3})^2 / (200 \times 10^{-6}) = 2.2 \text{ W/K}$. $G_{\text{rad}} \approx 3.5 \times 10^{-3} \text{ W/K}$ (same estimate as WE-07.1). $\Gamma = 0.0016 \ll 0.1$. **Conduction-dominated.** Select graded via pattern (Pattern 3, Table 07.7).

Step 2: Gaussian profile design (DR-09.5). Collection radius: $R_m = 10 \text{ mm}$ (half-substrate width). $\sigma = R_m / \sqrt{2} = 7.07 \text{ mm}$. Peak fill fraction at center: $f_{via}^{\text{max}} = 0.20$.

Five discrete zones:

Zone	r range (mm)	f_{via}	$\kappa_{\perp}^{\text{eff}}$ (W/mK)
1 (center)	0–2	0.200	81.1
2	2–4	0.165	67.1
3	4–6	0.110	45.1
4	6–8	0.058	24.4
5 (outer)	8–10	0.030	13.1

All zones use Cu fill ($\kappa_{via} = 401 \text{ W/mK}$), $d_{via} = 40 \mu\text{m}$. AR check: $200/40 = 5 < 10$. **DR-09.3: Pass.**

Step 3: CTE stress check. $\sigma_{via} = 130/0.65 \times (17 - 3.25) \times 10^{-6} \times 200 = 550 \text{ MPa}$ (Cu, $\Delta T = 200 \text{ K}$). Exceeds glass fracture limit at peak zone. **Mitigation:** W fill for zones 1–2 ($\kappa_{via} = 174 \text{ W/mK}$); Cu–W composite for zones 3–5.

Step 4: Network verification. Effective average $\kappa_{\perp}^{\text{eff}}$ across concentrator: $\approx 20 \text{ W/mK}$ (area-weighted). $G_{via} = 20 \times (20 \times 10^{-3})^2 / (200 \times 10^{-6}) = 40 \text{ W/K}$. $\Delta T_{\text{sub}} = 2/40 = 0.05 \text{ K}$. **Well within 3 K target.**

Step 5: EMT validity check. Zone 1 ($f_{via} = 0.20$): $V_{th} = 4 \text{ mm}/p_{via}$. For $d_{via} = 40 \mu\text{m}$ at $f_{via} = 0.20$: $p_{via} = 40 \sqrt{\pi/(4 \times 0.20)} = 79 \mu\text{m}$. $V_{th} = 4000/79 = 51 > 10$. Level 1 valid. $f_{via} = 0.20 < 0.30$: within MG validity. **Pass.**

Output: Via-Array Prescription Table

Parameter	Symbol	Value	DR
Fill material	—	W (zones 1–2), Cu–W (zones 3–5)	—
Via diameter	d_{via}	40 μm	DR-09.3: Pass
Pattern type	—	Gaussian-graded (5 zones)	DR-09.5
f_{via} range	—	0.030–0.200	DR-09.2: Pass
$\kappa_{\perp}^{\text{eff}}$ range	—	13–81 W/mK	Eq. (07.10)
Network G	G_{via}	40 W/K	Eq. (07.17)
CTE stress (max)	σ_{via}	0.14 MPa (W zones)	Pass
AR check	—	5.0	≤ 10 : Pass
EMT validity	V_{th}	≥ 51	Level 1: Valid
Overall	Pass	—	All DRs

07.8.3 Worked Example WE-07.3: ESC Keep-Out Zone Penalty

Problem: An electrostatic chuck uses a bipolar electrode pattern with electrode width $d_{\text{electrode}} = 5$ mm and clearance $d_{\text{clearance}} = 200$ μm , on a 5 mm-thick AlN chuck body ($\kappa = 170$ W/mK). Target: $G_{\text{chuck}} = 100$ W/K over a 300 mm wafer. Determine the required via fill fraction accounting for the KOZ penalty.

Step 1: KOZ geometry. For a bipolar pattern with alternating 5 mm electrodes and 0.4 mm gaps ($2 \times d_{\text{clearance}}$): $f_{\text{KOZ}} = 0.4/(5 + 0.4) \approx 0.074$.

Step 2: Required f_{via} without KOZ. $G_{\text{target}} = 100$ W/K over $A = \pi(150 \times 10^{-3})^2 = 0.0707$ m². Required $\kappa_{\perp}^{\text{eff}} = G \times t_{\text{sub}}/A = 100 \times 5 \times 10^{-3}/0.0707 = 7.07$ W/mK. Since $\kappa_{\text{matrix}} = 170$ W/mK for AlN, the bare substrate already provides $G_{\text{bare}} = 170 \times 0.0707/0.005 = 2,404$ W/K $\gg 100$ W/K.

Conclusion: For AlN, TGVs are not needed for bulk thermal conductance. The KOZ penalty is irrelevant because the base material already exceeds the target by $> 20\times$. TGVs in AlN would be used only for *anisotropy control* (e.g., directing heat to specific cooling channels) or for *electrical* interconnects, not for thermal conductance enhancement.

Mode-Path Matrix insight: This result illustrates DR-T.3 (critical path identification). For an AlN ESC, the bottleneck is *not* the chuck body but the wafer–chuck contact interface and the coolant channels. Design effort should target those edges, not the AlN bulk.

Counter-example: If the chuck body were borosilicate glass ($\kappa_{\text{matrix}} = 1.1$ W/mK): $G_{\text{bare}} = 1.1 \times 0.0707/0.005 = 15.5$ W/K. Need $G = 100$ W/K, so $\kappa_{\perp}^{\text{eff}} = 7.07$ W/mK. From Eq. (07.10) with Cu fill: $f_{\text{via}} = (7.07 - 1.1)/(401 - 1.1) = 0.015$. With KOZ penalty: $f_{\text{via}}^{\text{non-KOZ}} = 0.015/0.926 = 0.016$. Modest correction. **DR-09.7: Pass.**

Chapter Summary

This chapter established the complete pipeline from metamaterial theory to production-ready planar thermal devices:

1. **Cartesian translation** maps cylindrical cloaks/concentrators to rectangular substrates, with a formal cloaking efficiency metric R_{cloak} and the “Cartesian tax” (DR-09.10): typically 7–12% degradation, quantifiable and manageable through corner rounding, shell extension, and guard-ring vias.
2. **Switch plates** provide binary or continuous thermal conductance modulation, with realistic switching ratios of 100–500 (gas-gap) or 50–200 (mechanical contact) after accounting for radiation leakage, contact resistance, and cycling degradation.
3. **Via array EMT** gives analytical expressions for $\kappa_{\perp}^{\text{eff}}$ (exact parallel mixing) and $\kappa_{\parallel}^{\text{eff}}$ (Maxwell–Garnett, valid for $f_{\text{via}} < 0.30$), with a model fidelity ladder for escalation to FEA when validity boundaries are exceeded.
4. **Pattern selection** uses a decision table keyed to MPM bottleneck type, choosing among uniform, clustered, graded, and annular patterns.
5. **Glass substrate engineering** addresses the $100\times$ conductivity gap through TGV design rules (DR-09.1 through DR-09.9), CTE stress management, and fill material selection.
6. **ESC integration** handles the thermal-electrical co-design through keep-out zone analysis and via fill fraction penalty calculations.
7. **MPM integration** places via arrays as high-conductance edges in the thermal network, with a standard prescription table as the output format.

Part II is now closed. The reader has the complete conduction toolkit: transformation theory (Chapter 5), switchable devices (Chapter 6), and planar/via-array fabrication (this chapter). Every conduction-dominated path ($\Gamma < 0.1$) identified by the Mode-Path Matrix can now be addressed with a specific design tool and validated through a standard prescription output.

Table 07.8: Complete set of via/substrate design rules (DR-09 series, defined in Chapter 7).

Rule	Section	Description
DR-09.1	§07.5.1	TGV necessity for glass substrates: mandatory above 73 mW/cm^2 for $300 \text{ }\mu\text{m}$ borosilicate.
DR-09.2	§07.3.2	Minimum via fill fraction: $f_{\text{via}} > 0.03$ for meaningful thermal engineering.
DR-09.3	§07.5.2	TGV aspect ratio limit: $\text{AR} \leq 10$ for production reliability.
DR-09.4	§07.4.2	Hotspot via clustering: cluster footprint $> 1.2\times$ hotspot area.
DR-09.5	§07.4.3	Graded array Gaussian profile: $\sigma = R_m/\sqrt{2}$; ≥ 5 zones.
DR-09.6	§07.4.4	Annular cloak ring count: $N \geq 5$ rings.
DR-09.7	§07.6.2	ESC via placement: no vias in KOZ; compensate fill fraction.
DR-09.8	§07.5.4	Glass temperature limits: respect strain point.
DR-09.9	§07.5.4	Via-to-edge spacing: $d_{\text{edge}} \geq \max(3d_{\text{via}}, 300 \text{ }\mu\text{m})$.
DR-09.10	§07.1.2	Cartesian translation penalty: $\leq 10\%$ degradation from symmetry breaking; 3–5% additional for discrete lattice.

Looking Forward

This chapter concludes Part II: the Diffusionics toolkit for conduction-dominated paths ($\Gamma < 0.1$). Chapters 5–7 have provided the complete conduction design catalog: transformation thermotics (Ch. 5), switchable devices (Ch. 6), and via array architectures (this chapter).

Part III (Chapters 8–10) now turns to the complementary toolkit: Thermal Photonics for radiation-dominated paths ($\Gamma > 10$). Chapter 8 introduces spectral engineering and thermal window design, where the emissivity ε replaces the conductivity κ as the primary design variable.

Research Roadmap

Table 07.9: Research roadmap for Chapter 7 topics.

Topic	Current State	5-Year Horizon	TPD Connection
High-AR TGV	$\text{AR} \leq 10$ (production)	$\text{AR} \leq 20$ via advanced laser + ALD	Enables thicker glass interposers
Graded via arrays	Staircase (5 zones)	Continuous grading via maskless lithography	Approaches ideal GRIN thermal lens
Non-conductive thermal vias	AlN, diamond-like carbon	CVD diamond fill ($\kappa > 1000 \text{ W/mK}$)	ESC integration without KOZ penalty
Inverse design of via patterns	Manual pattern selection	Topology optimization of $f_{\text{via}}(\mathbf{x})$	Automated MPM-to-prescription pipeline
Active via switching	Fixed fill fraction	Phase-change fill (VO_2) for switchable vias	Merges Ch. 6 doublers with Ch. 7 vias

Problems

- 7.1. Cartesian cloak design.** A rectangular thermal cloak must protect a $4 \times 4 \text{ mm}^2$ region on a Si substrate. The outer frame extends to $12 \times 12 \text{ mm}^2$. (a) Compute the required anisotropy ratio from Eq. (07.4). (b) Can this be achieved with a binary Si/SiO₂ composite (Chapter 5)? (c) What via fill fraction would achieve the same anisotropy using Cu TSVs? (d) Estimate R_{cloak} including the Cartesian tax (DR-09.10).
- 7.2. Cartesian Cloak Design (2).** Design a Cartesian thermal cloak for a $4 \times 4 \text{ mm}^2$ square object embedded in a $20 \times 20 \text{ mm}^2$ silicon substrate ($t_{\text{sub}} = 500 \text{ }\mu\text{m}$). (a) Compute the required anisotropy ratio from Eq. (07.4). (b) Select a material pair from Table 05.5 that achieves this anisotropy. (c) Calculate the cloaking efficiency R_{cloak} degradation due to the Cartesian approximation using DR-09.10. (d) Propose a via array implementation and specify f_{via} , d_{via} , and p_{via} . (e) Fill in the via-array prescription table.
- 7.3. Via Array EMT Verification.** For a Cu-in-glass via array with $d_{\text{via}} = 40 \text{ }\mu\text{m}$, $p_{\text{via}} = 120 \text{ }\mu\text{m}$ (hexagonal lattice): (a) Calculate the fill fraction f_{via} . (b) Compute $\kappa_{\perp}^{\text{eff}}$ and $\kappa_{\parallel}^{\text{eff}}$ using Eqs. (07.10)–(07.11). (c) Include the Kapitza resistance correction (Eq. 07.13) with $R_K = 5 \times 10^{-8} \text{ m}^2 \text{ K/W}$ and recompute $\kappa_{\perp}^{\text{eff}}$. By what fraction does the anisotropy change? (d) Verify the EMT validity criteria V1–V5 for this array applied to a $5 \times 5 \text{ mm}^2$ hotspot. Compare your EMT result against Figure 07.4.
- 7.4. EMT validity assessment.** A clustered via array has $d_{\text{via}} = 30 \text{ }\mu\text{m}$, $p_{\text{via}} = 60 \text{ }\mu\text{m}$ ($f_{\text{via}} = 0.196$) in a $2 \times 2 \text{ mm}^2$ hotspot region on a glass substrate. (a) Check all five EMT validity criteria (V1–V5). (b) Which fidelity level (Table 07.5) is appropriate? (c) Estimate the expected EMT error using Figure 07.4.
- 7.5. Glass Interposer Thermal Budget.** A fan-out panel-level package uses a $500 \text{ }\mu\text{m}$ borosilicate glass substrate ($30 \times 30 \text{ mm}^2$) carrying a processor dissipating 15 W . (a) Verify DR-09.1: is the heat flux above the TGV necessity threshold? (b) Determine the minimum f_{via} (Cu fill) to limit ΔT_{sub} to 3 K . (c) Check DR-09.3 for $t_{\text{sub}} = 500 \text{ }\mu\text{m}$. (d) Evaluate the CTE mismatch stress (Eq. 07.21) for $\Delta T = 250 \text{ K}$ and assess cracking risk. (e) Propose a W-fill alternative and recalculate. (f) Fill in the prescription table for both designs.
- 7.6. Glass interposer thermal budget.** A glass interposer ($30 \times 30 \text{ mm}^2$, $400 \text{ }\mu\text{m}$ thick, $\kappa_{\text{matrix}} = 1.2 \text{ W/mK}$) carries four dies, each dissipating 3 W . (a) Compute Γ to confirm conduction dominance. (b) Design a uniform TGV array to keep $\Delta T < 5 \text{ K}$ per die. (c) Fill in the prescription table. (d) Check CTE stress for both Cu and W fill options.
- 7.7. Hotspot Via Cluster Design.** A $3 \times 3 \text{ mm}^2$ hotspot on a glass substrate dissipates 2 W . (a) Using DR-09.4, determine the cluster footprint. (b) Design the via cluster (d_{via} , p_{via} , f_{via}) to limit the local ΔT to 3 K through a $200 \text{ }\mu\text{m}$ substrate. (c) Construct the local Mode-Path Matrix and verify $\Gamma < 0.1$. (d) Check conductance matching (M) with a TIM interface ($G_{\text{TIM}} = 20 \text{ W/K}$). (e) Verify EMT validity criterion V3 (V_{th}).
- 7.8. KOZ penalty analysis.** An ESC with a quadrupolar electrode pattern has $f_{\text{KOZ}} = 0.15$. The target effective fill fraction is $f_{\text{via}} = 0.08$. (a) Compute the required local fill fraction in the non-KOZ region. (b) Verify EMT validity at this higher fill fraction. (c) At what f_{KOZ} would the compensated f_{via} exceed the MG validity limit of 0.30 ?
- 7.9. ESC Co-Design Challenge.** An electrostatic chuck requires a $200 \text{ }\mu\text{m}$ AlN dielectric

layer (electrode width 5 mm, clearance 200 μm). The chuck body is 5 mm-thick AlN ($\kappa = 170 \text{ W/mK}$). (a) Calculate the keep-out zone fraction f_{KOZ} for a bipolar electrode pattern. (b) Determine the effective via fill fraction needed in the non-KOZ regions to achieve $G = 100 \text{ W/K}$ over a 300 mm wafer (Eq. ??). (c) Compare interdigitated and peripheral frame layouts. Which provides better thermal uniformity?

7.10. Graded concentrator optimization. Redesign WE-07.2 with 7 zones instead of 5. (a) Compute the f_{via} values for each zone using the Gaussian profile (Eq. 07.18). (b) Compare the area-weighted $\kappa_{\perp}^{\text{eff}}$ to the 5-zone design. (c) Is the improvement worth the additional lithographic complexity?

7.11. Switch plate design. A wafer transport system requires $G_{\text{ON}}/G_{\text{OFF}} > 200$ at 350 K for a $200 \times 200 \text{ mm}^2$ plate. (a) Evaluate gas-gap and mechanical-contact options, accounting for the non-idealities. (b) Which design meets the requirement? (c) What surface treatment is needed for the OFF-state radiation leakage limit?

7.12. Network Integration. A 3D IC package consists of: die ($P = 10 \text{ W}$) $\rightarrow$ TSV interposer (Si, 300 μm) $\rightarrow$ glass panel (300 μm) $\rightarrow$ TIM $\rightarrow$ heat sink. (a) Construct the thermal network graph. (b) Build the Mode-Path Matrix. (c) Identify the critical path. (d) If the glass panel is the bottleneck, design a TGV array to relieve it. (e) After the improvement, does the critical path shift? If so, to which element? (f) Document using the prescription table format.

7.13. MPM network update. Starting from the 2.5D IC package network of WE-4.1 (Chapter 4), replace the substrate edge (originally $G = 30 \text{ W/K}$ for Si) with a glass substrate ($\kappa_{\text{matrix}} = 1.1 \text{ W/mK}$) plus a TGV array from WE-07.1. (a) Recompute the critical path and bottleneck ratio β . (b) Has the bottleneck moved? (c) What is the new system ΔT for 80 W total dissipation?

7.14. Multi-Objective Optimization. A glass interposer requires simultaneous thermal ($G > 5 \text{ W/K}$) and electrical (signal via density $> 100/\text{mm}^2$) performance. The thermal and electrical vias share the same substrate. (a) Determine whether a single via array can serve both functions (consider Cu fill for both). (b) If not, partition the via area into thermal and electrical zones and optimize f_{via} for each. (c) What is the minimum substrate area needed to satisfy both requirements simultaneously?

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